Yash Rawal

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Master of Computer Applications (AI & ML) | Microsoft Certified: Azure AI Fundamentals (AI-900)

Cover Letter: AI/ML Engineer — ECS Big Bets & AI 10.0 (Req ID: 2333072)

Optum (UnitedHealth Group)   Hyderabad, India   September 3, 2026

Dear Optum Hiring Team,

I am writing to express my enthusiastic interest in the AI/ML Engineer position within the ECS Big Bets and AI 10.0 initiatives at Optum (UnitedHealth Group) in Hyderabad (Req ID: 2333072).

As an Applied AI & Machine Learning Engineer currently architecting enterprise agentic platforms serving 1M+ active users across 120+ organizations, my background brings together the exact technical pillars required for Optum's next-generation health optimization: Model Context Protocol (MCP) server development, multi-agentic orchestration (LangChain, LangGraph), hierarchical RAG architectures, and clinical NLP experience.

- **Model Context Protocol (MCP) & Multi-Agent Systems,** Architected and deployed enterprise multi-agent workflows utilizing MCP Server Tool Routing (Tool RAG) and A2A communication, designing a capability contract layer that cut execution overhead by 3x and token consumption by 70%.
- **Clinical NLP & Healthcare AI Ingestion,** Engineered clinical summarization pipelines for 116K+ longitudinal medical records using 4-bit quantized Llama 3, and designed real-time WebSocket streaming Indic ASR with NeMo RNN-T and CTC decoding.
- **Hierarchical RAG & Vector Knowledge Retrieval,** Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a 90%+ Recall@10 retrieval rate.
- **Evaluation Control Planes & Guardrail Security,** Built an automated LLM evaluation and red-teaming control plane to score agent reasoning trajectories, and secured agentic workflows against OWASP LLM vulnerabilities with phantom-token context isolation and automated PII redaction.

Delivering care and connecting millions of patients with clinical insights through AI 10.0 requires high-concurrency systems engineered with strict privacy compliance, deterministic tool invocation, and low-latency agentic execution. Having built production agentic platforms at 1M+ scale, contributed upstream to Spring AI (PR #5711), and engineered clinical NLP pipelines, I am eager to help Optum advance health optimization on a global scale.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my technical expertise in MCP, multi-agentic architectures, and clinical AI can contribute to Optum's ECS Big Bets initiatives.

Sincerely,
Yash Rawal